openwec

openwec

install ubuntu 22

```
apt  install libclang-dev libkrb5-dev  libgssapi-krb5-2 sqlite msktutil
adduser openwec
cargo build --release
cp ./target/release/openwecd /usr/local/bin/
cp ./target/release/openwec /usr/local/bin/
systemctl edit openwec.service --full --force
```

compatible avec systemd

```
### openwec.service
[Unit]
Description=Windows Events Collector
After=network.target

[Service]
Type=simple
User=openwec
Restart=always
RestartSec=5s
ExecStart=/usr/local/bin/openwecd -c  /etc/openwec/openwec.conf.toml

[Install]
WantedBy=multi-user.target
```

droits avec systemd

```
chown -R openwec.openwec /var/db/openwec
```

faire de meme pour les directory de sortie des logs

logs systemd

```
journalctl -u openwec.service -f
```

config openwec

```
# /etc/openwec.conf.toml par default
[server]
verbosity = "info"
db_sync_interval = 5
flush_heartbeats_interval = 5

[database]
type = "SQLite"
# You need to create /var/db/openwec yourself
path = "/var/db/openwec/db.sqlite"

[[collectors]]
hostname = "openwec.jmp.local"
listen_address = "0.0.0.0"

[collectors.authentication]
type = "Kerberos"
service_principal_name = "http/openwec.jmp.local@JMP.LOCAL"
keytab = "/etc/krb5.keytab"
```

Openwec s'attend à trouver son fichier de conf sous /etc

```
root@ubuntu22:~# ln -s /etc/openwec/openwec.conf.toml /etc/openwec.conf.toml
```

init database

```
openwec -c /etc/openwec/openwec.conf.toml db init
```

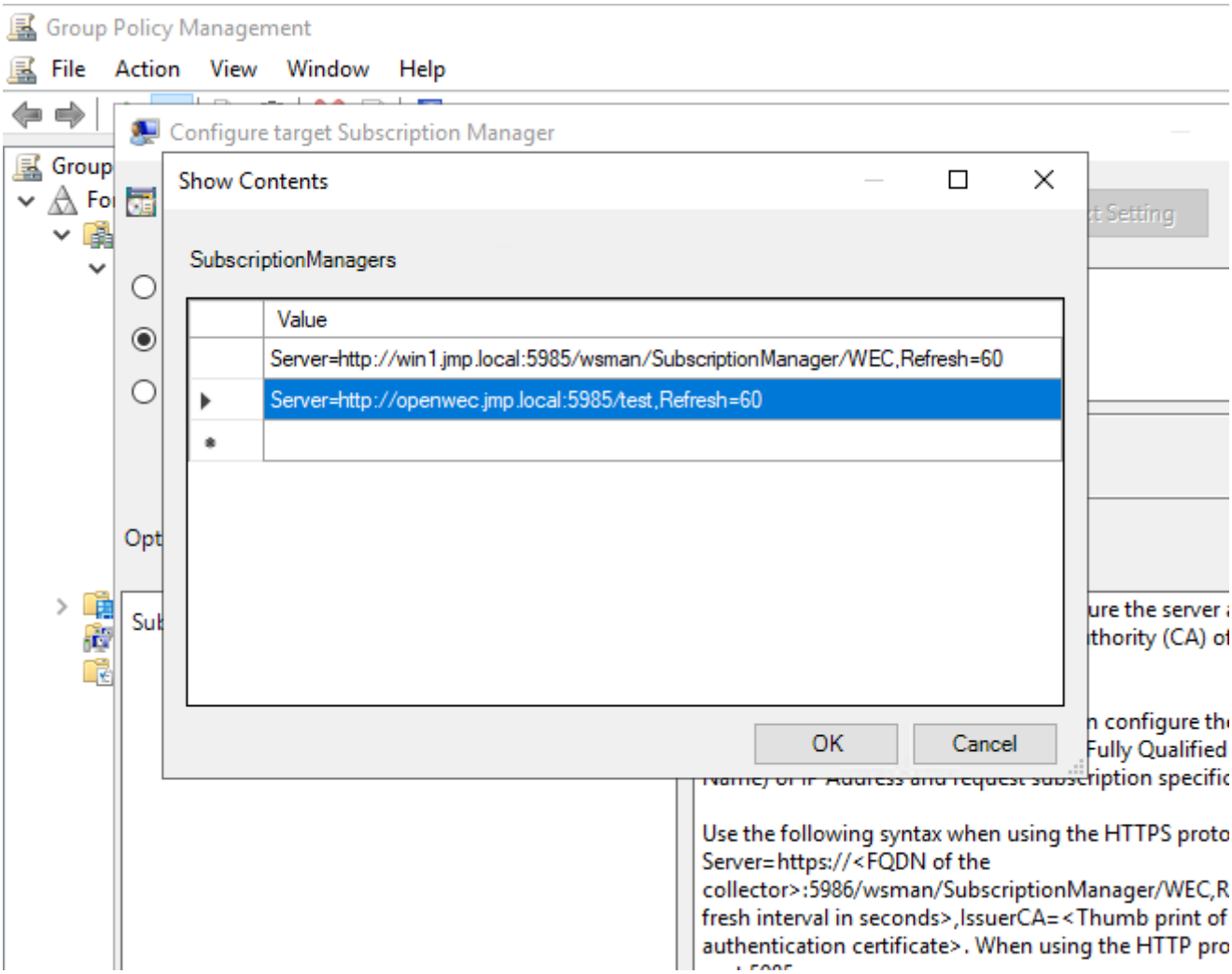
config de openwec avec le xml de l'ANSSI

```
wget https://raw.githubusercontent.com/ANSSI-FR/guide-journalisation-microsoft/main/Standard_WEC_query.xml

openwec -c /etc/openwec/openwec.conf.toml subscriptions new anssi-subscription ./ansi.xml
Subscription anssi-subscription has been created successfully. You need to configure its outputs using `openwec subscriptions edit anssi-subscription outputs add --help`. When you are ready, you can enable it using `openwec subscriptions edit anssi-subscription --enable

openwec subscriptions edit  anssi-subscription outputs add --format json files /openwec/logssho an
openwec subscriptions enable anssi-subscription
```

reprendrela GPO pour windows et rajouter le server openwec

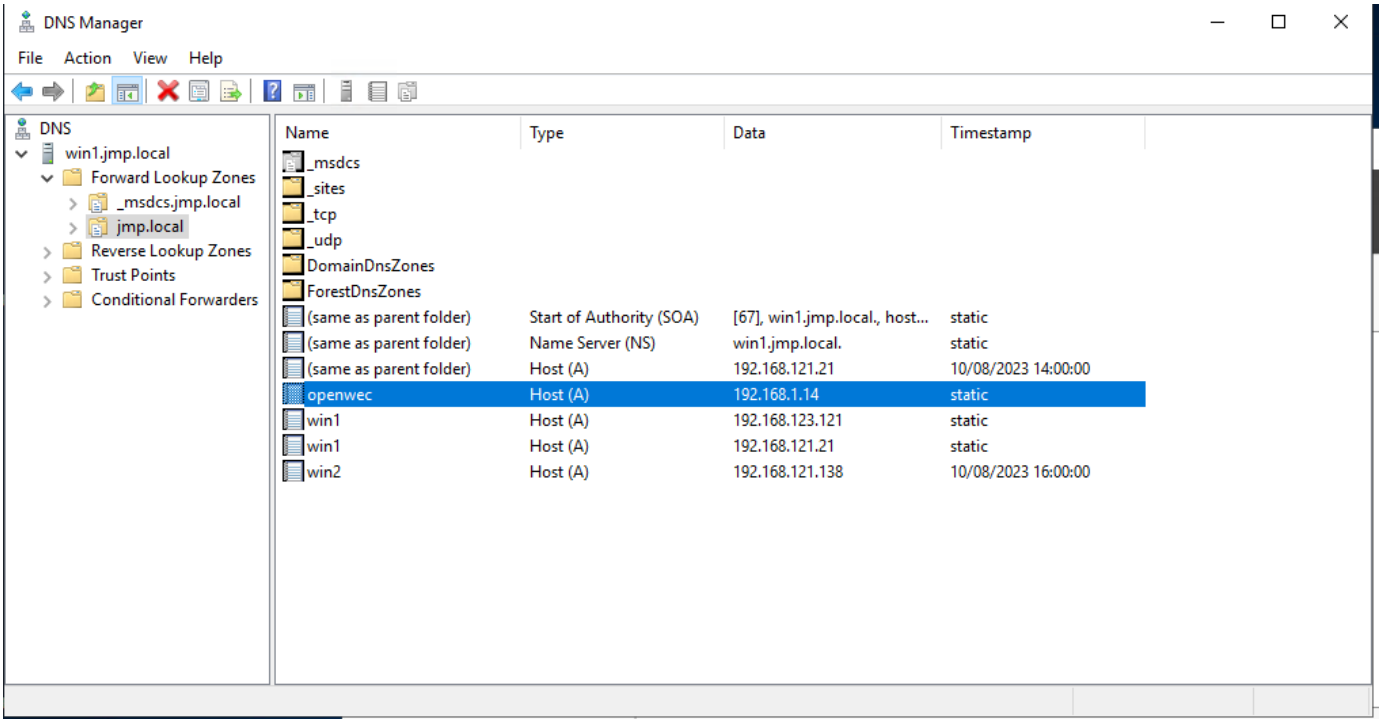


kerberos

modifiez /etc/systemd/resolved pour y mettre le dns windows
et (le forwading ne marche pas?) /etc/hosts

192.168.121.21 win1.jmp.local
192.168.121.138 win2.jmp.local
192.168.26.14 openwec.jmp

Mettre l'IP au dns de windows

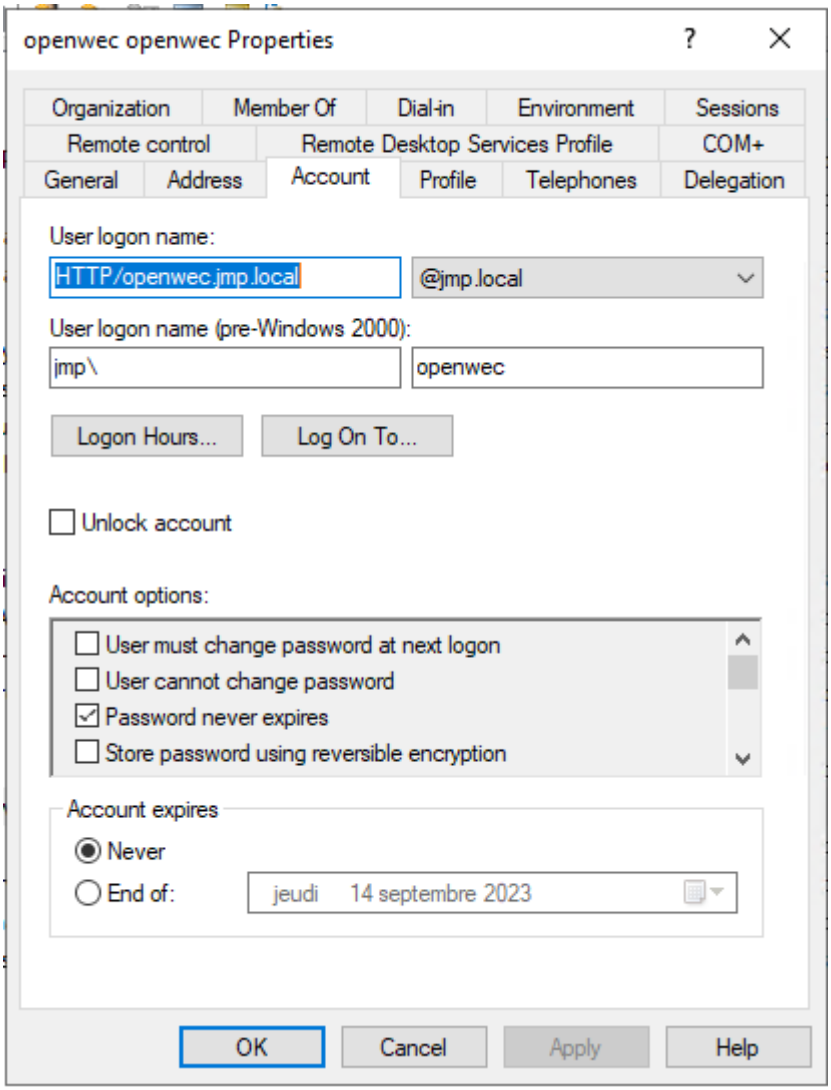


Attention à la date heure pour que kerberos fonctionne ntp sur la machine openwec

Pas besoin de mettre la machine openwec sur le domaine
il suffit de créer un utilisateur openwec, relier le SPN , génerer le keytab sur le DC et le transférer sur la machine openwec

sur le DC créer l'utilsateur openwec avec la console AD

pas sur que loginname soit nécessaire de cette façon mais ca marche
au final ne rien toucher d'autres y compris sur l'encryption



On a besoin de relier le SPN à l'utilisateur OpenWec

```
setspn -S HTTP/openwec.jmp.local openwec
## list spn pour un compte
setspn -L openwec
Registered ServicePrincipalNames for CN=openwec
openwec,CN=Users,DC=jmp,DC=local:
        HTTP/openwec.jmp.local
```

génération du keytab sur le DC

```
jmp\vagrant@WIN1 C:\Users\vagrant>ktpass -princ
HTTP/openwec.jmp.local@JMP.LOCAL -mapuser openwec -crypto ALL -mapop set -
ptype KRB5_NT_PRINCIPAL -pass jSaliutdb#03 -target win1.jmp.local -kvno 0 -
out c:\Users\vagrant\Desktop\openwec.keytab
Using legacy password setting method
Failed to set property 'servicePrincipalName' to 'HTTP/openwec.jmp.local' on
Dn 'CN=openwec,CN=Users,DC=jmp,DC=local': 0x13.
WARNING: Unable to set SPN mapping data.
If openwec already has an SPN mapping installed for HTTP/openwec.jmp.local,
this is no cause for concern.
```

```
Key created.
Key created.
Key created.
Key created.
Key created.
Output keytab to c:\Users\vagrant\Desktop\openwec.keytab:
Keytab version: 0x502
keysize 59 HTTP/openwec.jmp.local@JMP.LOCAL ptype 1 (KRB5_NT_PRINCIPAL) vno
7 etype 0x1 (DES-CBC-CRC) keylength 8 (0x7319abfd38a291fb)
keysize 59 HTTP/openwec.jmp.local@JMP.LOCAL ptype 1 (KRB5_NT_PRINCIPAL) vno
7 etype 0x3 (DES-CBC-MD5) keylength 8 (0x7319abfd38a291fb)
keysize 67 HTTP/openwec.jmp.local@JMP.LOCAL ptype 1 (KRB5_NT_PRINCIPAL) vno
7 etype 0x17 (RC4-HMAC) keylength 16 (0x2b0e4a20181d10cec58c1b725201f1cc)
keysize 83 HTTP/openwec.jmp.local@JMP.LOCAL ptype 1 (KRB5_NT_PRINCIPAL) vno
7 etype 0x12 (AES256-SHA1) keylength 32
(0x3eb1313fb66506d404b8e70f59f1aae7180a8deb9a784ff2a7b30555976b838d)
keysize 67 HTTP/openwec.jmp.local@JMP.LOCAL ptype 1 (KRB5_NT_PRINCIPAL) vno
7 etype 0x11 (AES128-SHA1) keylength 16 (0x46374d22595f53ca680ce1a9b389fc1f)
```

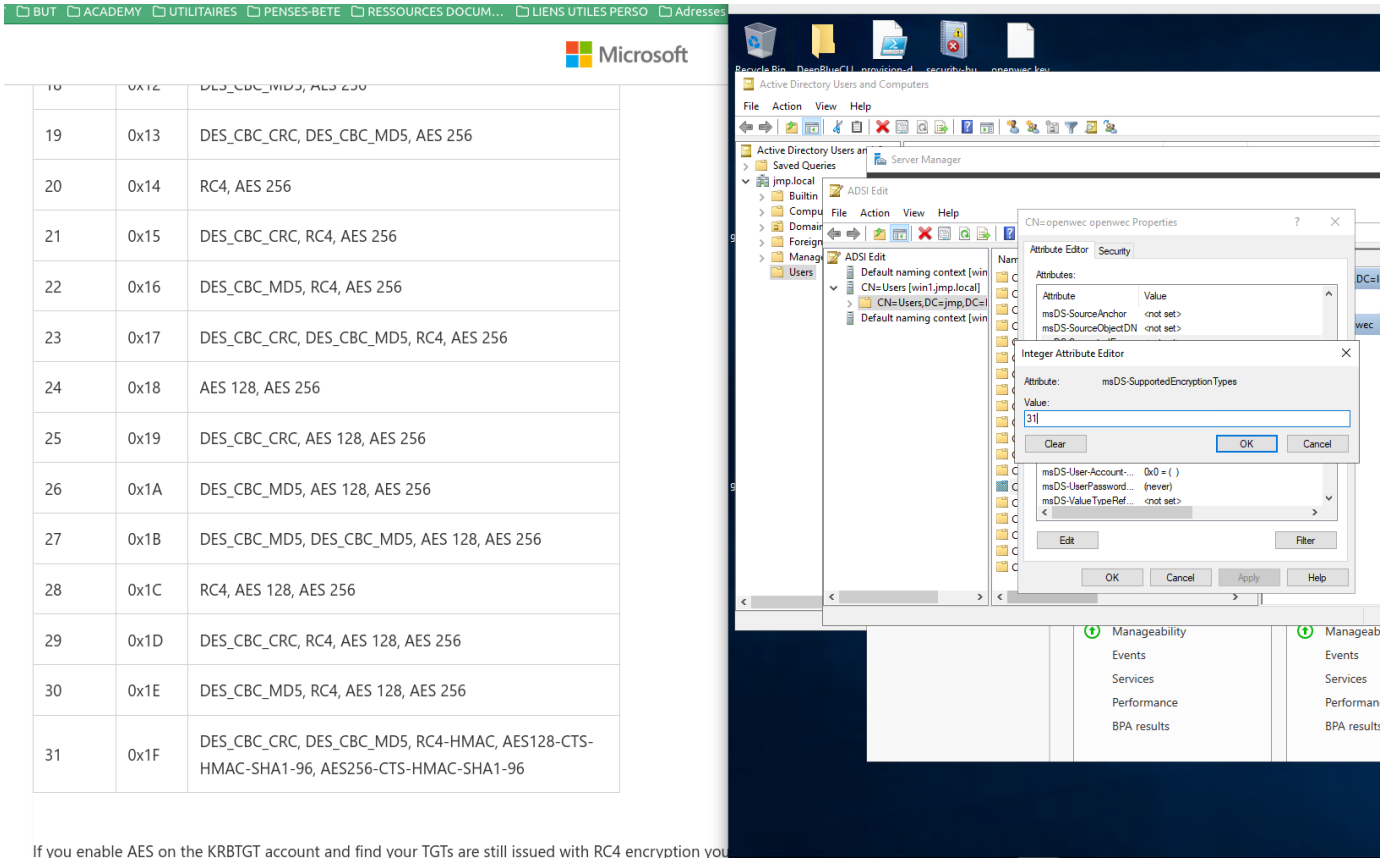
on recopie ensuite le keytab sur la machine

On peut tester sans démarrer systemd

```
/usr/local/bin/openwecd -c  /etc/openwec/openwec.conf.toml
```

non utile mais pour memo

KRB5_TRACE=/dev/stdout kinit -kt /etc/openwec.keytab HTTP/openwec.jmp.local@JMP.LOCAL



19	0x13	DES_CBC_CRC, DES_CBC_MD5, AES 256
20	0x14	RC4, AES 256
21	0x15	DES_CBC_CRC, RC4, AES 256
22	0x16	DES_CBC_MD5, RC4, AES 256
23	0x17	DES_CBC_CRC, DES_CBC_MD5, RC4, AES 256
24	0x18	AES 128, AES 256
25	0x19	DES_CBC_CRC, AES 128, AES 256
26	0x1A	DES_CBC_MD5, AES 128, AES 256
27	0x1B	DES_CBC_MD5, DES_CBC_MD5, AES 128, AES 256
28	0x1C	RC4, AES 128, AES 256
29	0x1D	DES_CBC_CRC, RC4, AES 128, AES 256
30	0x1E	DES_CBC_MD5, RC4, AES 128, AES 256
31	0x1F	DES_CBC_CRC, DES_CBC_MD5, RC4-HMAC, AES128-CTS-HMAC-SHA1-96, AES256-CTS-HMAC-SHA1-96

If you enable AES on the KRBTGT account and find your TGTs are still issued with RC4 encryption you

si l'heure est bien mise sur les machines

double nat ?

```
août 14 23:45:57 openwec.jmp.local openwecd[3747]: [DEBUG server] Received HTTP request from 192.168.1.111:49928: POST /test
août 14 23:45:57 openwec.jmp.local openwecd[3747]: [DEBUG server] Successfully authenticated WIN1$@JMP.LOCAL
août 14 23:45:57 openwec.jmp.local openwecd[3747]: [DEBUG server::multipart] Header found: Header { name: "Content-Type", value: "application/HTTP-Kerberos-session-encrypted" }
août 14 23:45:57 openwec.jmp.local openwecd[3747]: [DEBUG server::multipart] Header found: Header { name: "OriginalContent", value: "type=application/soap+xml;charset=UTF-16;Length=1434" }
août 14 23:45:57 openwec.jmp.local openwecd[3747]: [DEBUG server::multipart] Header found: Header { name: "Content-Type", value: "application/octet-stream" }
août 14 23:45:57 openwec.jmp.local openwecd[3747]: [DEBUG server::kerberos] Try to decrypt Kerberos payload
août 14 23:45:57 openwec.jmp.local openwecd[3747]: [DEBUG server::kerberos] Kerberos payload decrypted successfully
août 14 23:45:57 openwec.jmp.local openwecd[3747]: [DEBUG server::logic] Received http://schemas.microsoft.com/wbem/wsman/1/wsman/End request
août 14 23:45:57 openwec.jmp.local openwecd[3747]: [INFO  server] Responded status 200 OK to 192.168.1.111:49928 (request was POST:/test) in 0.397ms
août 14 23:45:58 openwec.jmp.local openwecd[3747]: [DEBUG server::subscription] Update subscriptions from db (interval tick)
août 14 23:45:58 openwec.jmp.local openwecd[3747]: [DEBUG server::subscription] Active subscriptions are: {"7D0BCB08-F229-4534-A3E3-A34F0D77EC06"}
août 14 23:45:58 openwec.jmp.local openwecd[3747]: [INFO  server::subscription] Active subscriptions are: "anssi-subscription" (A7E43988-C5C0-4E2D-95B3-3459A334CFFB)
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server] Received HTTP request from 192.168.1.111:49917: POST /wsman/subscriptions/7D0BCB08-F229-4534-A3E3-A34F0D77EC06
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server] Successfully authenticated WIN1$@JMP.LOCAL
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::multipart] Header found: Header { name: "Content-Type", value: "application/HTTP-Kerberos-session-encrypted" }
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::multipart] Header found: Header { name: "OriginalContent", value: "type=application/soap+xml;charset=UTF-16;Length=22764" }
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::multipart] Header found: Header { name: "Content-Type", value: "application/octet-stream" }
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::kerberos] Try to decrypt Kerberos payload
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::kerberos] Kerberos payload decrypted successfully
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::sldc] Try to decompress SLDC data (22764 compressed bytes)
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::sldc] SLDC decompression succeed (73568 uncompressed bytes)
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::logic] Received http://schemas.dmtf.org/wbem/wsman/1/wsman/Events request
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [INFO  server::logic] Received Events from 192.168.1.111:49917 (WIN1$@JMP.LOCAL) for subscription anssi-subscription (A7E43988-C5C0-4E2D-95B3
-3459A334CFFB)
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::outputs::file] Computed path is /openwec/logs/192.168.1.111/WIN1@JMP.LOCAL/messages
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::outputs::file] File /openwec/logs/192.168.1.111/WIN1@JMP.LOCAL/messages is already opened
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::outputs::file] Computed path is /data/logs/192.168.1.111/WIN1@JMP.LOCAL/messages
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::outputs::file] File /data/logs/192.168.1.111/WIN1@JMP.LOCAL/messages is already opened
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::logic] Store bookmark from 192.168.1.111:49917 (WIN1$@JMP.LOCAL) for subscription anssi-subscription (A7E43988-C5C0-4E2D-95B3-
3459A334CFFB): <BookmarkList><Bookmark Channel="Application" RecordId="1676"/><Bookmark Channel="Microsoft-Windows-AppLocker/EXE and DLL" RecordId="0"/><Bookmark Channel="Microsoft-Windows-App
Locker/MSI and Script" RecordId="0"/><Bookmark Channel="Microsoft-Windows-AppLocker/Packaged app-Deployment" RecordId="0"/><Bookmark Channel="Microsoft-Windows-AppLocker/Packaged app-Execution
" RecordId="0"/><Bookmark Channel="Microsoft-Windows-DriverFrameworks-UserMode/Operational" RecordId="27"/><Bookmark Channel="Microsoft-Windows-EventCollector/Operational" RecordId="0"/><Bookm
ark Channel="Microsoft-Windows-Forwarding/Operational" RecordId="368"/><Bookmark Channel="Microsoft-Windows-LSA/Operational" RecordId="1"/><Bookmark Channel="Microsoft-Windows-PowerShell/Opera
tional" RecordId="41099"/><Bookmark Channel="Microsoft-Windows-SmartCard-Audit/Authentication" RecordId="0"/><Bookmark Channel="Microsoft-Windows-Sysmon/Operational" RecordId="38564"/><Bookmar
k Channel="Microsoft-Windows-TaskScheduler/Operational" RecordId="5538"/><Bookmark Channel="Microsoft-Windows-TerminalServices-RDPClient/Operational" RecordId="0"/><Bookmark Channel="Microsoft
-Windows-Windows Defender/Operational" RecordId="231"/><Bookmark Channel="Microsoft-Windows-WinRM/Operational" RecordId="45900" IsCurrent="true"/><Bookmark Channel="Security" RecordId="1323671
"/><Bookmark Channel="System" RecordId="9789"/><Bookmark Channel="Windows PowerShell" RecordId="32261"/></BookmarkList>
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::heartbeat] Store event heartbeat for WIN1$@JMP.LOCAL (192.168.1.111) with subscription A7E43988-C5C0-4E2D-95B3-3459A334CFFB
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [INFO  server] Responded status 200 OK to 192.168.1.111:49917 (request was POST:/wsman/subscriptions/7D0BCB08-F229-4534-A3E3-A34F0D77EC06) in
 14.285ms
août 14 23:45:59 openwec.jmp.local openwecd[3747]: [DEBUG server::heartbeat] Cache heartbeat for WIN1$@JMP.LOCAL (192.168.1.111) with subscription A7E43988-C5C0-4E2D-95B3-3459A334CFFB. last_se
en = 1692049559, last_event_seen = Some(1692049559)
août 14 23:46:03 openwec.jmp.local openwecd[3747]: [DEBUG server::subscription] Update subscriptions from db (interval tick)
août 14 23:46:03 openwec.jmp.local openwecd[3747]: [INFO  server::heartbeat] Flush heartbeat cache
août 14 23:46:03 openwec.jmp.local openwecd[3747]: [DEBUG server::subscription] Active subscriptions are: {"7D0BCB08-F229-4534-A3E3-A34F0D77EC06"}
août 14 23:46:03 openwec.jmp.local openwecd[3747]: [INFO  server::subscription] Active subscriptions are: "anssi-subscription" (A7E43988-C5C0-4E2D-95B3-3459A334CFFB)
août 14 23:46:03 openwec.jmp.local openwecd[3747]: [INFO  server::heartbeat] Heartbeat cache flushed and cleared
```

The following lines are desirable for IPv6 capable hosts

::1 ip6-localhost ip6-loopback

```
[Resolve]
DNS=192.168.121.21
#FallbackDNS=8.8.8.8
```

```
#Domains=iutbeziers.fr
LLMNR=no
MulticastDNS=no
DNSSEC=no
#DNSOverTLS=no
#Cache=yes
#DNSStubListener=yes
#ReadEtcHosts=yes
```

```
 ping win1.jmp.local
PING win1.jmp.local (192.168.121.75) 56(84) bytes of data.
^C
--- win1.jmp.local ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1027ms
```

```
[libdefaults]
        default_realm = JMP.LOCAL

# The following krb5.conf variables are only for MIT Kerberos.
        kdc_timesync = 1
        ccache_type = 4
        forwardable = true
        proxiable = true

# The following encryption type specification will be used by MIT Kerberos
# if uncommented.  In general, the defaults in the MIT Kerberos code are
# correct and overriding these specifications only serves to disable new
# encryption types as they are added, creating interoperability problems.
#
# The only time when you might need to uncomment these lines and change
# the enctypes is if you have local software that will break on ticket
# caches containing ticket encryption types it doesn't know about (such as
# old versions of Sun Java).

#       default_tgs_enctypes = des3-hmac-sha1
#       default_tkt_enctypes = des3-hmac-sha1
#       permitted_enctypes = des3-hmac-sha1

# The following libdefaults parameters are only for Heimdal Kerberos.
        fcc-mit-ticketflags = true

[realms]
        JMP.LOCAL = {
                kdc = win1.jmp.local
```

```
                admin_server = win1.jmp.local
        }
[domain_realm]
        .jmp.local = JMP.LOCAL
        jmp.local = JMP.LOCAL
        .mit.edu = ATHENA.MIT.EDU
        mit.edu = ATHENA.MIT.EDU
        .media.mit.edu = MEDIA-LAB.MIT.EDU
        media.mit.edu = MEDIA-LAB.MIT.EDU
        .csail.mit.edu = CSAIL.MIT.EDU
        csail.mit.edu = CSAIL.MIT.EDU
        .whoi.edu = ATHENA.MIT.EDU
        whoi.edu = ATHENA.MIT.EDU
        .stanford.edu = stanford.edu
        .slac.stanford.edu = SLAC.STANFORD.EDU
        .toronto.edu = UTORONTO.CA
        .utoronto.ca = UTORONTO.CA
```

```
root@ubuntu22:~/openwec# kinit vagrant
Password for vagrant@JMP.LOCAL:
root@ubuntu22:~/openwec# klist
Ticket cache: FILE:/tmp/krb5cc_0
Default principal: vagrant@JMP.LOCAL

Valid starting       Expires              Service principal
13/08/2023 18:41:30  14/08/2023 04:41:30  krbtgt/JMP.LOCAL@JMP.LOCAL
        renew until 14/08/2023 18:41:27
root@ubuntu22:~/openwec#
```

```
root@ubuntu22:~/openwec# msktutil create --computer-name openwec --user-
creds-only --server win1.jmp.local --no-reverse-lookups    No computer
account for openwec found, creating a new one.
root@ubuntu22:~/openwec# klist -ke
Keytab name: FILE:/etc/krb5.keytab
KVNO Principal
---- --------------------------------------------------------------------
---
   1 openwec$@JMP.LOCAL (DEPRECATED:arcfour-hmac)
   1 openwec$@JMP.LOCAL (aes128-cts-hmac-sha1-96)
   1 openwec$@JMP.LOCAL (aes256-cts-hmac-sha1-96)
   1 OPENWEC$@JMP.LOCAL (DEPRECATED:arcfour-hmac)
   1 OPENWEC$@JMP.LOCAL (aes128-cts-hmac-sha1-96)
   1 OPENWEC$@JMP.LOCAL (aes256-cts-hmac-sha1-96)
   1 host/ubuntu22@JMP.LOCAL (DEPRECATED:arcfour-hmac)
```

```
   1 host/ubuntu22@JMP.LOCAL (aes128-cts-hmac-sha1-96)
   1 host/ubuntu22@JMP.LOCAL (aes256-cts-hmac-sha1-96)
root@ubuntu22:~/openwec#

root@ubuntu22:~/openwec# systemctl restart openwec
root@ubuntu22:~/openwec# systemctl status openwec
● openwec.service - Windows Events Collector
     Loaded: loaded (/etc/systemd/system/openwec.service; disabled; vendor
preset: enabled)
     Active: active (running) since Sun 2023-08-13 16:51:07 CEST; 19ms ago
   Main PID: 68603 ((openwecd))
      Tasks: 1 (limit: 9232)
     Memory: 448.0K
        CPU: 5ms
     CGroup: /system.slice/openwec.service
             └─68603 "[openwecd]"

août 13 16:51:07 ubuntu22 systemd[1]: Started Windows Events Collector.
```

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@openwec:~# sudo realm -v discover win1.jmp.local

- Resolving: _ldap._tcp.win1.jmp.local
- Resolving: win1.jmp.local
- Performing LDAP DSE lookup on: 192.168.121.21
- Successfully discovered: jmp.local
  jmp.local
  type: kerberos
  realm-name: JMP.LOCAL
  domain-name: jmp.local
  configured: no
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
  root@openwec:~#